Membrane Air Separation

Pre-Lab Plan
Unit Operations Lab # 1
09/17/2025
Group #3, Tuesday 9 AM

Anna Cai, Hussein Rezk, Adil Feratovic, Aaron Casas

1. Experimental Objective (5 pts)

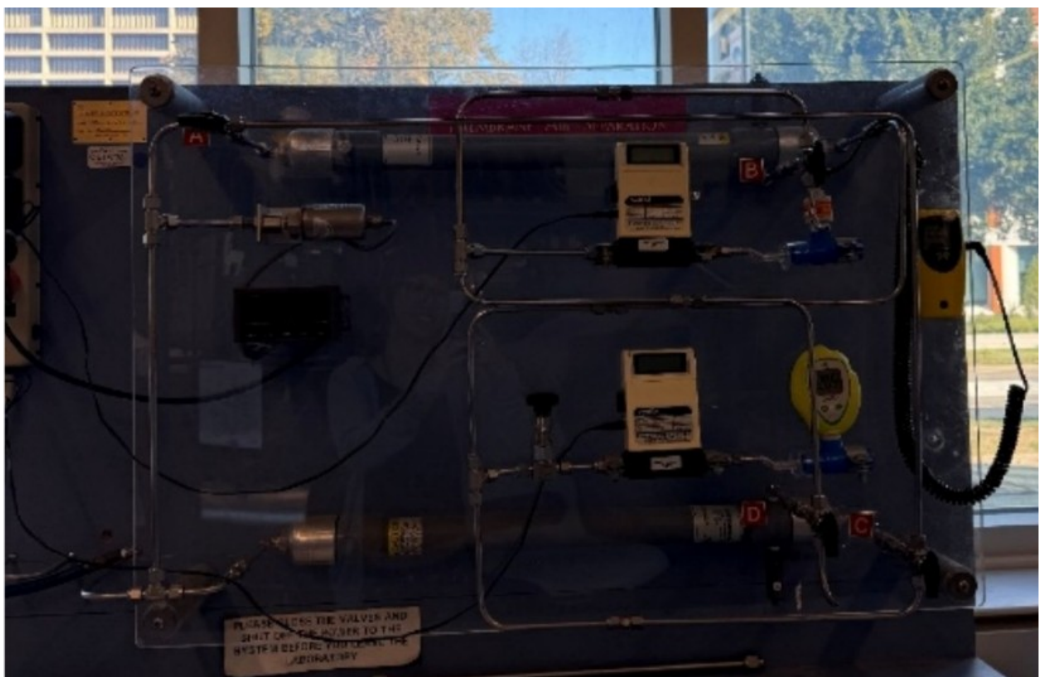
Characterizing the pressure-driven permeation of O_2/N_2 by using measured flow rates and oxygen percentages from a hollow fiber membrane air separation rig. We will look at two configurations, parallel and series, and compare performances across transport coefficients and separation factors. We will then plot these as functions of operating pressure, with tube-side gas flow rate and parallel and series as parameters.

We will then analyze and draw conclusions from the results. Discuss the trends with pressure, rate, and configuration.

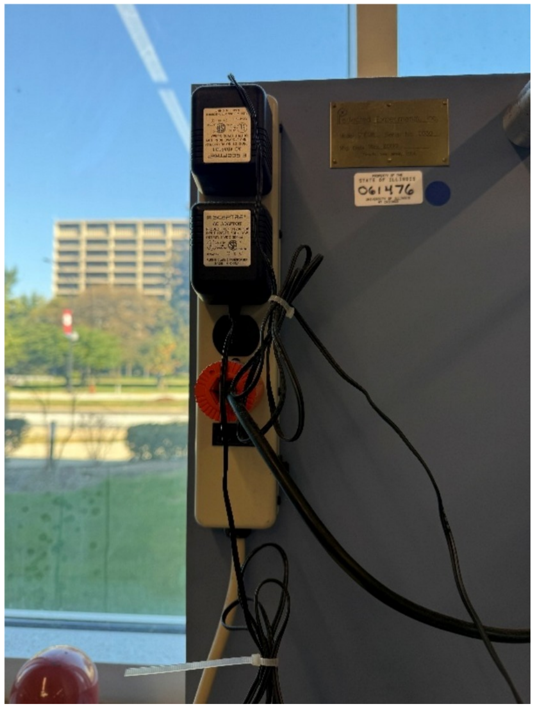
2. Experimental

This experiment uses 2 membranes packed with hollow tubes, where O2 diffuses through the fiber walls. Air from the cylinder passes through regulators, filters, and flow meters. The valve system allows the operations of parallel or series to be conducted with the total flow rate added amongst the tube and shell sides.

2.1 Apparatus (5 pts)



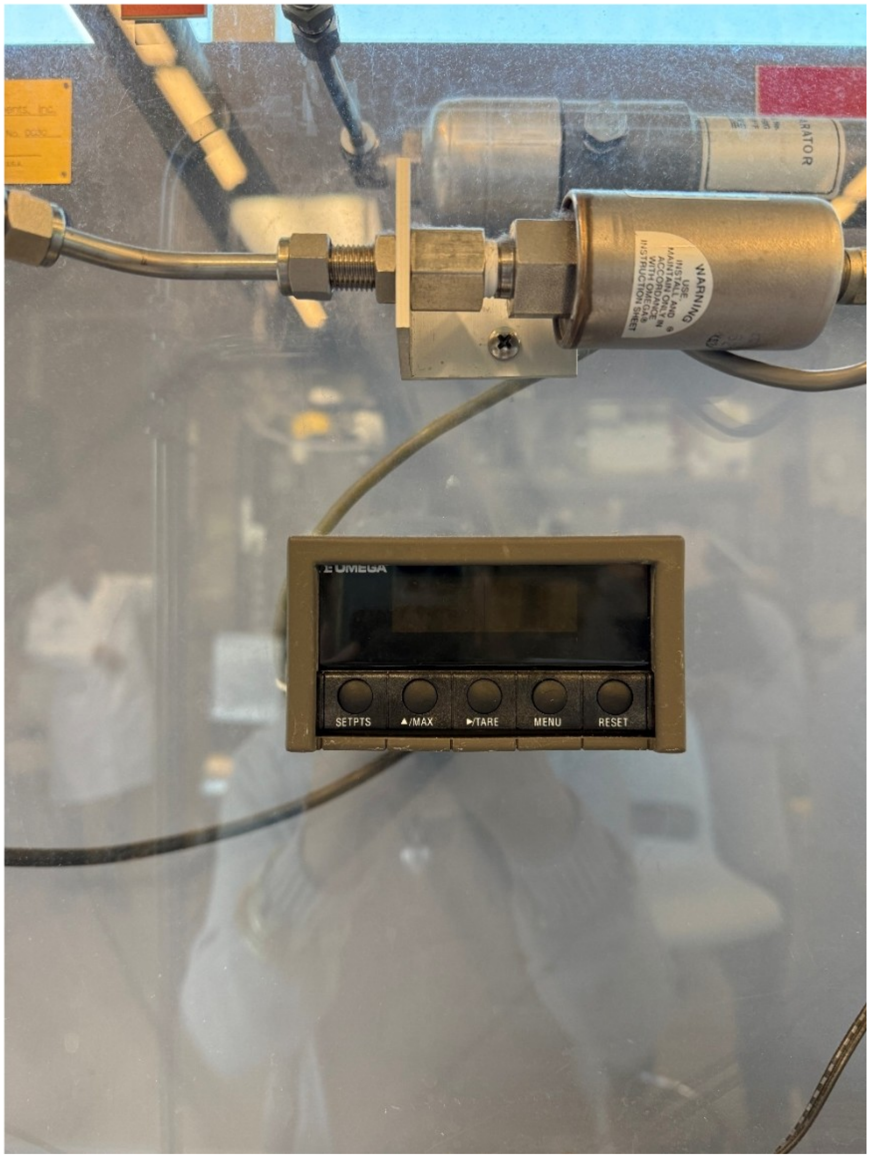
The experimental apparatus consists of two separator columns that can be operated in series or in parallel. The modules contain hollow fibers that are non-porous, semi-permeable membranes. The two columns are mounted together on an acrylic board. Each column has $2.7 m^2$ of membrane area, they are 1 inch in diameter and three feet long.



This is an electrical strip that supplies power to the pressure and O_2 gauges.

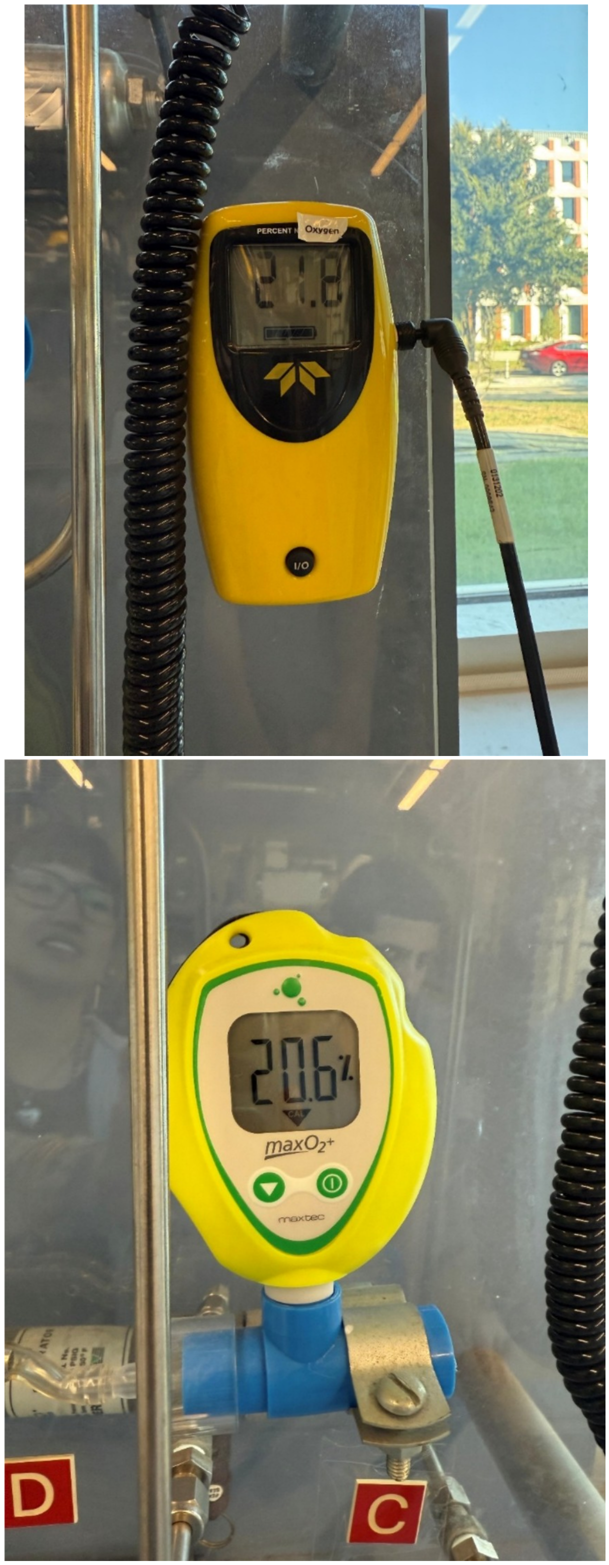
High pressure air tank.

Regulator and pass-through valves on the air tank.



Pressure gauge

Mass flow meters



O_2 gauges.

2.2 Materials and Supplies (5 pts)

Equipment	Description/Usage
Oxygen Sensor	Reads oxygen levels of the pipe
Valves	Allows for configurations (Parallel/Series)
Flow rate controller	Allows for the user to control the rate
Air Tank	Allows for air pressure to be adjusted in the pipes
Pressure gauge	Reads pressure pass through into the pipes

2.3 Experimental Plan

Procedure:
- Turn on both oxygen analyzers and wait for 10 minutes. Calibrate to 20.9% O_2. Make sure flow meters and pressure display are on
- Open the compressed air and take note of cylinder pressure. Use the regulator to set run pressures
- Set up the parallel test. Series will be done after
- Adjust the needle valve to the desired tube side effluent flowrate. Recheck pressure changes
- Once steady, record pressure, flow rates, and O_2 %s
- There will be around 15 runs done for parallel and series configuration each

Ranges:
Pressure will range from 50 to 180 PSIG
$0 \le O_2\% \le 100$

Variables:
Independent – Pressure, flowrates, and module configuration
Dependent – O_2 %, K_{O2}, K_{N2}, O_2/N_2 recovery, and α

Expectations:
O_2 % higher on shell side than feed.

Replicates:
If time and materials permit, 1 set of replicated data will be collected for parallel and series

Fits:

Plot permeation rate vs ΔP and look at slope and R^2
Plot α′ vs pressure and stage cut. Look at the trends
Compare experimental α′/ α ′ ′ to α inferred from K

Data Reduction/Calculations:
Convert to easier units
Compute partial pressures to find K_j
Compute α, α′, α′ ′, and stage cut
Plot K_{O2}, K_{N2} vs pressure

Theory Links:
Use solution diffusion trends and definitions, such as permeability and selectivity to interpret trends
Evaluate whether K_j is pressure independent or not

Project Safety Evaluation

The project safety evaluation on the next page is an analysis of the potential safety hazards in the experiment, and ways of controlling these hazards. The process parameters used in the experiment, such as temperature and pressure ranges, are also listed. When listing hazards, be specific on the conditions; how and where they may occur, and ranges of values for the hazard (pressure, temperature, concentration). The MSDS for all chemicals should be reviewed, especially hazardous and toxic chemicals, but also include materials like water, and compressed air.

Think about how the hazards in the experiment are controlled and describe them in detail. For instance, using a fume hood to handle a toxic chemical is a safety control that reduces your exposure to the chemical.

Project Safety Evaluation (10 pts)

	Yes/No	
Potential electrical hazards?	Yes	If yes, identify conditions: The potential electrical hazard is poor connections that can lead to short circuits, while making sure everything is grounded.
Potential mechanical hazards?	no	If yes, identify conditions:
Potential pressure hazards?	yes	If yes, identify conditions: The pressurized tank can potentially bounce around and injure people if not properly clamped to the workbench.

Potential temperature hazards?	No	If yes, identify conditions:
Hazardous/toxic chemicals involved?	No	If yes, hazards involved:
Gloves required?	No	If yes, specify type:
MSDS reviewed by all team members?	Yes	List MSDS sheets reviewed: Compressed air
Process Parameters	*Max Expected*	*List location and possible hazards and precautions*
Temperature (ºC)	N/A	
Pressure (psia)	1200psig	The pressurized tank is located to the left of the workbench. A pressurized tank can potentially bounce around and injure people if not properly clamped to the workbench. Another Hazard is that if heat is present around the pressurized tank, it can potentially blow up.
Safety Controls	*yes/no*	*If yes, Provide description*
Valves and surge protectors.	*Yes*	*We have three valves: the pressure pass through, the pressure regulator, and the pressure release valve. These valves allow us to control the pressure being fed into the apparatus. The surge protectors allow us to easily switch off the power if there's an electrical issue with the apparatus itself.*

I have read relevant background material for the Unit Operations Laboratory entitled: "____" and understand the hazards associated with conducting this experiment. I have planned out my experimental work in accordance to standards and acceptable safety practices and will conduct all of my experimental work in a careful and safe manner. I will also be aware of my surroundings, my group members, and other lab students, and will look out for their safety as well.

Aaron Casas 9-17-25

date

Hussein Rezk 9-17-25

date

Anna Cai 9-17-25

date

9-17-25

Adil Feragovic date